

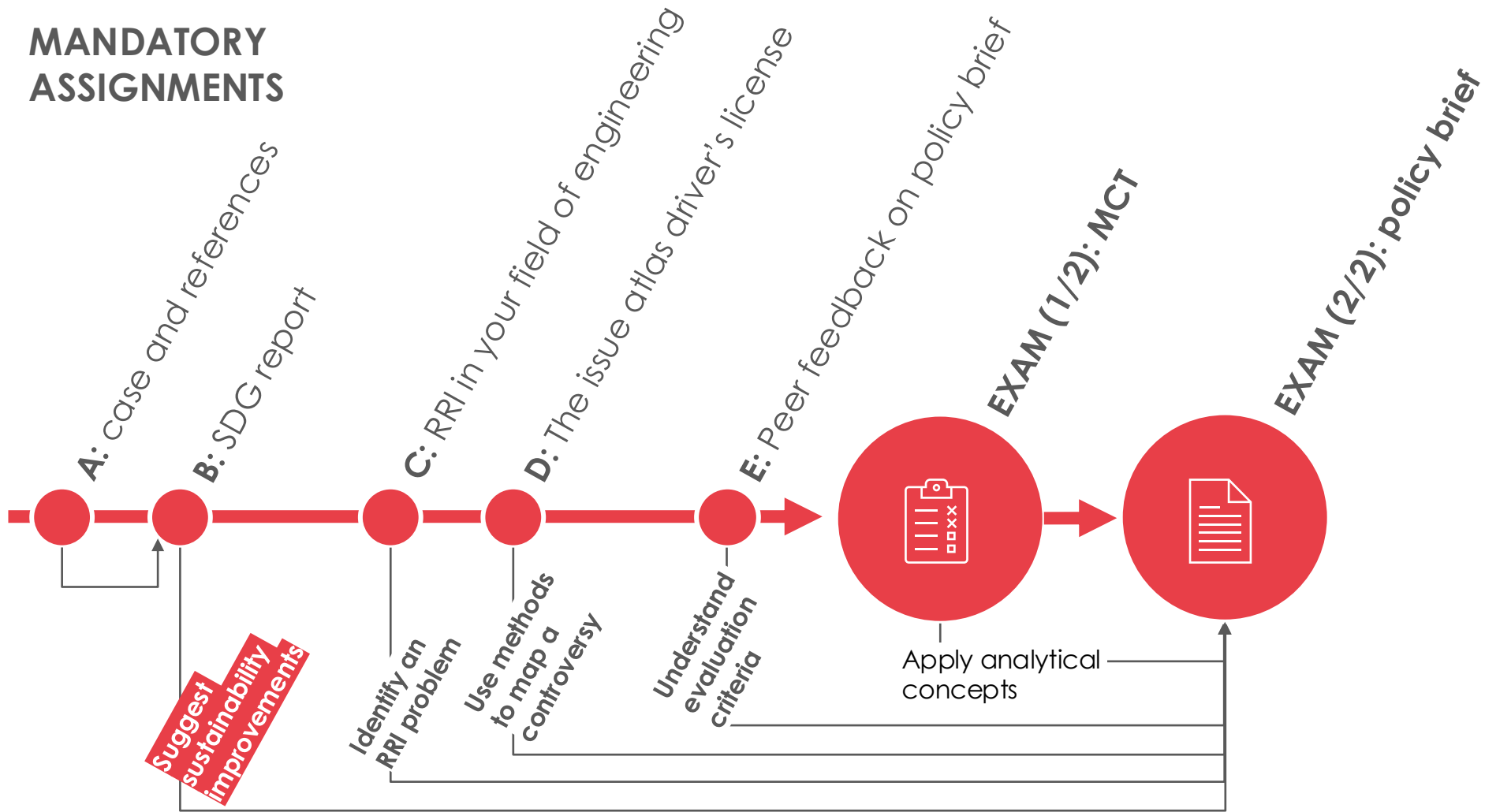
WELCOME TO PART 2

Thanks for your suggested sustainability improvements!

Now, how will they be received by real world actors in the controversy over the Danish energy islands?

And how are you going to handle it?

MANDATORY ASSIGNMENTS



Monday 12/1	Tuesday 13/1	Wednesday 14/1	Thursday 15/1	Friday 16/1
Responsible Research and Innovation (RRI)	Sociology of science and expertise	Controversy mapping introduction	Social acceptance	Markets in the making
Ethics	Risk and uncertainty	Mapping #1: Explore the issue atlas	Mapping #3: Making data-driven points	Universal design
The politics of technology	Mandatory assignment #C RRI in your field	Mapping #2: Follow the actors	Mapping #4: Visual storytelling	Mandatory assignment #D Issue atlas driver's license

Monday 19/1	Tuesday 20/1	Wednesday 21/1	Thursday 22/1	Friday 23/1
Users, innovation and resistance	Write policy brief	Mandatory assignment #F Policy conference	Write policy brief	MULTIPLE-CHOICE TEST
Public engagement	Write policy brief	Write policy brief	Write policy brief	Write policy brief
RRI recap and policy brief intro	Q&A based supervision	Q&A based supervision (online)	Write policy brief	HAND IN POLICY BRIEF

Learning objectives

- Establish the significance of SDGs in relation to engineering projects and discover how they can be used in practice and analysis.
- Know and understand prevailing definitions of sustainability and the history of the concept of sustainability.
- Understand the importance of adopting a life cycle/systems perspective and considering a comprehensive range of impacts within each of the three sustainability dimensions - environment, society and economy.
- **Identify how social values are embedded in engineering projects and how technology can become a matter of public concern.**
- **Describe the role of ethics in engineering work within one's own engineering field and in collaboration with other fields and stakeholders.**
- **Describe and evaluate the primary social processes of designing, implementing, adopting, and contesting innovation.**
- **Discuss key dimensions of responsible research and innovation and apply them to evaluate engineering projects, especially anticipation, reflexivity, inclusion, and responsiveness.**
- **Recommend responsible strategies to foster absolute social and environmental sustainability of engineering projects.**

PART 2

Assignments **C**, **D**, and **E**

RRI and Wicked Problems

42622 Science, Technology & Society

Benjamin Lipp | Anders Kristian Munk | Tanja Schneider

Science and Technology Studies | Department of Technology, Management and Economics

Part 2 teaching team from DTU Management



Benjamin

Responsible

All inquiries about part 2

bmili@dtu.dk



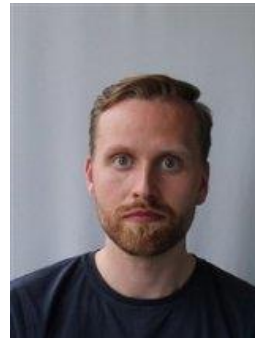
Mathieu



Anders



Tanja



Daniel

And our TAs
**Emma, Alfred,
Zalikhha, Jakob,
Sarah, Arantxa,
Lois, Isabela,
Emilie Frederik,
and Amelie +
invited guests**

Wicked problems

“The search for scientific bases for confronting problems of social policy is bound to fail, because of the nature of these problems. They are “wicked” problems, whereas science has developed to deal with “tame” problems. Policy problems cannot be definitively described. Moreover, **in a pluralistic society there is nothing like the undisputable public good; there is no objective definition of equity**; policies that respond to social problems cannot be meaningfully correct or false; and **it makes no sense to talk about “optimal solutions” to social problems unless severe qualifications are imposed** first. Even worse, there are no “solutions” in the sense of definitive and objective answers”

(Rittel & Webber, 1973)



Horst Rittel ▲

▼ Early operations research (WWII)



Factors of problem wickedness

Reduced list from the original 10 factors*

- ▶ There is no stopping rule. The problem is ongoing.
- ▶ Solutions are irreversible.
- ▶ The way the problem is described determines its possible solutions
- ▶ Problem can be described as a symptom of other problems
- ▶ Solutions can be better or worse, not true or false.

**Rittel, Horst WJ, and Melvin M. Webber. "Dilemmas in a general theory of planning." Policy sciences 4.2 (1973): 155-169.*

Exercise preview: assess the problem wickedness of the challenges below

- ▶ Protect Copenhagen from future sea level rise.
- ▶ Decrease vaccine hesitancy by combatting online disinformation through better science communication.
- ▶ Build a 4.3 meter high storm surge protection with vertical sheet piles around Lynetteholm to handle the currently predicted 100-year flood scenario for Copenhagen in 2070.
- ▶ Develop the next generation of mRNA vaccines.
- ▶ Make cities more resilient to climate change.
- ▶ Get commuters to bike, take public transport, or use car pooling services instead of their private car.
- ▶ Develop new metamodeling techniques that make it possible to run simulations on the national transport model 100 times faster.
- ▶ Mitigate the societal impact of future pandemics.

Problems have no stopping rule.

They are ongoing.

▶ Some solutions temporarily stabilize situations, but the underlying complexity of those situations mean they evolve and with them the nature of the problem.

Example: Efforts to alleviate poverty might temporarily work, but since poverty is a complex phenomenon with multiple interrelated causes, the reasons why people are poor constantly change, thus destabilizing the solution.

▶ Further, in most cases, you can always imagine an even better state.

Example: What does an end to poverty even look like? What are the criteria?

▶ Other solutions are more like interventions: they affect the field and, in turn, the nature of the problem. They have unintended consequences.

Example: Efforts to alleviate crime will have effects on the way criminals behave. They adjust and develop new strategies = the nature of the problem has changed as a consequence of the solution.

Solutions are irreversible.

Little room to experiment.

▶ One of the typical ways in which solutions become interventions is by creating path dependence.

Example: If electric cars are a solution to the climate crisis, their implementation require significant investment in charging infrastructure that will in turn make it very hard to experiment with other solution strategies.

▶ This means that it is hard to learn through low-cost trial-and-error.

▶ Further, even without path dependence, solutions to wicked problems can be a one-shot, high stakes operation.

Example: If new types of nuclear reactors are a solution to the climate crisis, the consequences of a failure could be catastrophic with consequences beyond NOT solving the climate crisis.

The way the problem is described determines its possible solutions

▶ Tame problems can have objective answers because they are well defined. Wicked problems are ill-defined leaving them open to interpretation.

▶ This means that multiple concerned groups with different values, different stakes, and different problem definitions end up in framing controversies.

Example: Should circumcision be allowed for infant boys? Is this a religious question? Is it a medical question about pain levels and health benefits/problems? Is it a legal question about the rights of the parent vs. the rights of the child?

▶ These differences in worldviews and problem frames means that different solutions are relevant and different experts needed to help develop them.

Problem can be described as a symptom of other wicked problems

▶ One of the main challenges of wicked problems being ill-defined is that it is hard to delimit them from other wicked problems.

Example: Mitigating unemployment and aligning the development of artificial intelligence with societal interests are now interdependent wicked problems.

▶ This adds to complexity - it is essentially yet another way in which the way a problem is described determines its possible solutions - and puts further responsibility on us to be explicit about assumptions and framings when trying to solve problems.

Solutions can be better or worse, not true or false.



Because of differing interests, solutions will be better for some and worse for others. The best we can hope to do is to be explicit about who they are good for and how.

Example: Gentrification of urban areas can be great for those who own property (real estate prices go up), for the municipality (tax revenues go up and houses get renovated), and for local businesses (buying power of inhabitants go up), but not so great for citizens who benefited from cheap housing and unused/less regulated urban spaces.

Consequence:
We need the responsible engineer

Responsible **R**esearch and **I**nnovation

A brief prehistory of RRI

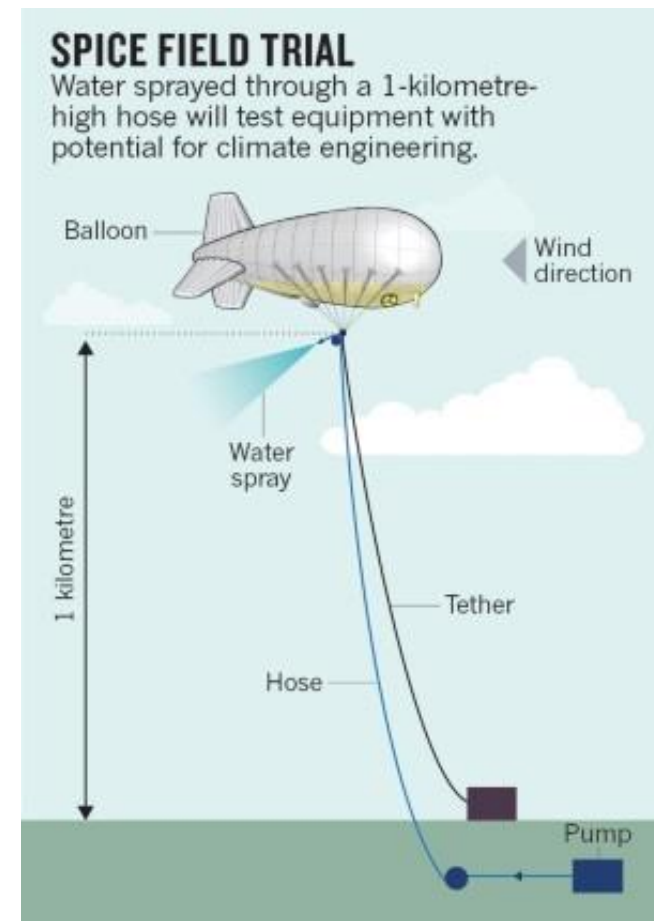
➤ History is full of examples of emerging technologies that turned out to have adverse consequences for humans and societies, although they promised great progress at their time of inception. Asbestos, Polychlorinated Biphenyls (PCBs), Social Media...

➤ Francis Bacon (17th century): science “for the relief of man’s estate”; J.D Bernal (1939): science in the service of society; Vannevar Bush (1945): the endless frontier of science.

➤ Societal push for science communication, public engagement, and mission driven research from 1980s

➤ Ideas pioneered in the latter half of the 20th century by researchers in the natural and technical sciences - e.g. synthetic biology, nuclear physics, artificial intelligence - concerned about dual use and the consequences of their innovations for humanity, society, environment.

➤ **Called RRI from the early 2000s onwards.**



SPICE project (Stratospheric Particle Injection for Climate Engineering) ▲

The **AREA** framework for RRI

Anticipate, Reflect, Engage, Act*

▶ Developed in the UK by the Engineering and Physical Sciences Research Council based on foundational work by sociologists of science and technology Jack Stilgoe, Richard Own and Phil Macnaghten in the late 00s.

▶ Four key dimensions:

*You have already done
some of this in Part 1*

But to a lesser extent this

- **Anticipate:** Explore possible impacts on society, environment, etc., intended and unintended, avoid restricting to what can be predicted.
- **Reflect:** Consider purpose of and motivation for research, assumptions, associated uncertainties, areas of ignorance, framings, etc.
- **Engage:** Open the above discussions on visions and impacts to broader deliberation through dialogue with relevant societal stakeholders.
- **Act:** Use these processes to influence the direction and trajectory of the research and innovation process itself. Why is that hard? How to do it?

**Originally anticipation, reflexivity, inclusion, and responsiveness in the foundational research policy paper by Stilgoe et al (2013)*

The **AREA** framework for RRI

Anticipate, Reflect, Engage, Act*

▶ Developed in the UK by the Engineering and Physical Sciences Research Council based on foundational work by sociologists of science and technology Jack Stilgoe, Richard Own and Phil Macnaghten in the late 00s.

▶ Four key dimensions:

*While mostly assuming
you know this*

- **Anticipate:** Explore possible impacts on society, environment, etc., intended and unintended, avoid restricting to what can be predicted.
- **Reflect:** Consider purpose of and motivation for research, assumptions, associated uncertainties, areas of ignorance, framings, etc.

*In Part 2, we will focus
more on these aspects*

- **Engage:** Open the above discussions on visions and impacts to broader deliberation through dialogue with relevant societal stakeholders.
- **Act:** Use these processes to influence the direction and trajectory of the research and innovation process itself. Why is that hard? How to do it?

**Originally anticipation, reflexivity, inclusion, and responsiveness in the foundational research policy paper by Stilgoe et al (2013)*

Expected result:
The responsible engineer

Examples of RRI requirements in practice

- ▶ Between 2014 and 2020, RRI was an explicit requirement for projects funded under the European Commission's €80 billion research and innovation program H2020.
- ▶ The principles and lessons learned are now integrated in its €95 billion successor program Horizon Europe.
- ▶ In the UK, the Engineering and Physical Sciences Research Council (EPSRC) actively promotes RRI through their AREA (Anticipate, Reflect, Engage, Act) framework.
- ▶ Similarly, Innovation Fund Denmark implements the European Commission's definition of RRI in their evaluation criteria
- ▶ And the IEEE code of ethics promotes responsibility by committing to:
 - improve "societal implications of conventional and emerging technologies"
 - "comply with ethical design and sustainable development practices"
 - "hold paramount the safety, health, and welfare of the public"
 - "avoid conflicts of interest whenever possible, and to disclose them to affected parties when they do exist"

Anticipate

- ▶ Take uncertainty seriously: It is hard to predict the effects of emerging technologies.
- ▶ The goal is to explore/imagine as many **possible** intended and unintended consequences of the research/innovation as possible. **As opposed to** establishing with certainty the probability of only the consequences that we understand well.
- ▶ Typical impulse of science is to narrow the imagination around impacts to what falls within area of expertise. Anticipation must work against that impulse.
- ▶ Comfortable questions within area of expertise: do we have the appropriate methodology, risk assessment, data management plan, lab health and safety, ethical approval, informed consent?
- ▶ Expand to questions like:
 - is this socially desirable?
 - for which stakeholders?
 - in what scenarios?
 - addressing what kind of larger societal challenges?

Reflect

▶ Science is already good at reflecting - it is organized scepticism (Popper).

▶ In this context: reflect on the consequences of own assumptions, of the way we have chosen to frame the problem, of our motivations for doing the research.

▶ Expand to:

- Which groups or individuals have a stake in the research and its outcomes?
- Which groups or individuals influence the research? In what ways do these stakeholders bring their influence to bear?
- Which groups or individuals can be held responsible in any way for unwanted/harmful consequences of the research?

▶ How do you know the answer to these questions? Who decides who the right stakeholders are? How do we decide what kind of stakes they have and who should be held accountable?

Engage

- ▶ Invite societal stakeholders to participate in your anticipation and reflection.
- ▶ Do they begin from the same assumptions as you? Do they see the same purpose and motivation for doing the research as you? Do they frame the problems to be solved in the same way as you?
- ▶ Do they agree with you on who the relevant people or groups are to invite, i.e. do they agree on who has a stake?
- ▶ Do they agree on who should be held accountable? Do they agree with you on who has influence on the research?
- ▶ Can they imagine other intended or unintended consequences of the research/innovation than you? Can they imagine other scenarios?
- ▶ Do they agree with you on what is socially acceptable/desirable?

Act

How do we do this?

What will it require in practice?

▶ Interdisciplinary training and collaboration. Basically why you are doing this course.

▶ Methods for technology assessment, scenario scanning, public engagement and deliberation, co-design, etc. On this course you will learn to:

- Conduct sustainability assessment (Part 1)
- Map and analyse public controversies around technology and innovation (Part 2)
- Organize public engagement exercises (Part 2)
- Apply design methods with diverse stakeholders in mind (Part 2)

▶ Creating structural incentives. Refer back to the way the current funding system for research and innovation incorporates RRI principles.